

2016.0 RANGE ROVER (LG), 303-00

ENGINE SYSTEM – GENERAL INFORMATION

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the 3.0L Diesel engine, refer to the relevant Description and Operation section in the workshop manual.

REFER to: [Engine](#) (303-01A Engine - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Coolant level ▪ Coolant leaks ▪ Oil level ▪ Oil leaks ▪ Visibly damaged or worn parts ▪ Loose or missing nuts or bolts	▪ Wiring harness ▪ Electrical connector(s) ▪ Injectors ▪ Glow plugs ▪ 5 volt sensor supply ▪ Sensor(s) ▪ Cooling fan control module and motor

- Engine Control Module (ECM)

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the concern is not visually evident, verify the symptom and refer to the relevant Symptom Chart. Symptom Charts have been separated into **Leaks** and **Noise Vibration and Harshness (NVH)** for ease of use. Alternatively, check for Diagnostic Trouble Codes (DTCs) and refer to the relevant DTC Index.

SYMPTOM CHARTS

Symptom Chart, Leaks

SYMPTOM	POSSIBLE CAUSE	ACTION
External coolant leaks	▪ Damaged hose(s) ▪ Damaged expansion tank ▪ Damaged radiator ▪ Leaking seals/gaskets ▪ Cracked/damaged casings	▪ For cooling system checks, REFER to: Engine Cooling (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Diagnosis and Testing).
Internal coolant leaks Note: This may be indicated by the production of white smoke from the exhaust	▪ Leaking seals/gaskets ▪ Cracked/damaged casings	▪ For cooling system checks, REFER to: Engine Cooling (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Diagnosis and Testing).
Engine overheats	▪ Insufficient coolant ▪ Insufficient oil ▪ Pressure cap fault ▪ Thermostat not opening	▪ For cooling system checks, REFER to: Engine Cooling (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Diagnosis and Testing).

	▪ Coolant pump failure ▪ Cooling fan failure	
Engine takes too long to reach operating temperature	▪ Thermostat stuck open	▪ For cooling system checks, REFER to: Engine Cooling (303-03A Engine Cooling - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Diagnosis and Testing).
External oil leaks	▪ Gaskets ▪ Seals ▪ Oil pipes ▪ Oil filter ▪ Oil cooler ▪ Damaged/cracked casings ▪ Crankcase ventilation system ▪ Piston ring blow-by	▪ Clean and confirm the area of the leak. Check the visual condition of oil carrying components. Check the crankcase ventilation system. Carry out a compression test, GO to Pinpoint Test A.
Internal oil leaks (leaks into coolant or combustion chamber) Note: This may be indicated by the production of blue smoke from the exhaust	▪ Gaskets ▪ Seals ▪ Damaged/cracked casings ▪ Worn valve guides ▪ Worn cylinder bores/pistons ▪ Broken piston rings	▪ Check for traces of oil in the coolant. Check for evidence of oil in the combustion chambers (deposits on the glow plugs, etc). Confirm oil consumption and vehicle usage with the owner/driver. Carry out an oil consumption test, GO to Pinpoint Test B.

Symptom Chart, NVH

NOTE:

As the checks suggested here are open to interpretation, they should be used as a guide only. Descriptions of noises, etc, are in general terms, so depend on a degree of experience on the part of the technician.

SYMPTOM	POSSIBLE CAUSE	ACTION
Rattle/ticking from top of engine	- Valve gear noise - Camshaft bearing noise - Camshaft chain noise - Tensioner noise - Vacuum pump noise - High pressure fuel pump noise	Check the engine oil pressure, GO to Pinpoint Test C. . Check the function of the hydraulic tappets and the camshaft condition. Check the camshaft bearings, chains and tensioners. Check the vacuum pump, and high pressure fuel pump
Growl from top of engine	- High pressure fuel pump belt noise - High pressure fuel pump belt tensioner noise	Check the high pressure fuel pump belt and tensioner
Squeaking/Creaking/Squeal from front of engine	- Front End Accessory Drive (FEAD) belt - FEAD belt tensioner - Driven components on FEAD belt	Check the FEAD belt and driven components, REFER to: Accessory Drive (303-05A Accessory Drive - TDV6 3.0L Diesel, Diagnosis and Testing).
Whine/Slap/Growl from front of engine	- Front End Accessory Drive (FEAD) belt - FEAD belt tensioner	Check the FEAD belt and driven components. Check the timing belt and tensioners

	▪ Driven components on FEAD belt ▪ Timing belt noise ▪ Timing belt tensioner noise	
Knock from lower half of engine (often worse with a cold engine)	▪ Piston slap ▪ Piston pin noise ▪ Connecting rod bearing noise	▪ Check the engine oil pressure, GO to Pinpoint Test C. . Check piston, cylinder bore, piston pin and connecting rod bearing for excess wear
Knock/Rumble from lower half of engine (often worse on overrun)	▪ Main bearing noise	▪ Check the engine oil pressure, GO to Pinpoint Test C. . Check connecting rod bearing for excess wear
Misfire/Rough running	▪ Engine management system ▪ Fuel charging and controls ▪ Exhaust gas recirculation (EGR) system ▪ Burnt/sticking valves ▪ Worn valve guides ▪ Worn cylinder bores/pistons ▪ Broken piston rings ▪ Damaged/cracked casings	▪ For engine management system tests, REFER to: Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module (ECM) (100-00 General Information, Description and Operation). For fuel charging and controls systems tests, REFER to: Fuel Charging and Controls (303-04A Fuel Charging and Controls - TDV6 3.0L Diesel, Diagnosis and Testing), Turbocharger (303-04B Fuel Charging and Controls - Turbocharger - TDV6 3.0L Diesel, Diagnosis and Testing). For EGR system tests, REFER to: Engine Emission Control (303-08A Engine Emission Control - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Diagnosis and Testing). Carry out a compression test, GO to Pinpoint Test A. Check for excess wear in engine components

PINPOINT TESTS

NOTE:

Where reference is made to 'suitable equipment', this refers to standard workshop equipment. Refer to the operating instructions for your own equipment when performing any tests.

WARNING:

Only compression testers able to read the higher compression pressures found in diesel engines should be used. Failure to follow this instruction may result in personal injury.

NOTE:

Where possible, compression testing should be carried out on an engine at operating temperature.

CAUTION:

If engine oil is introduced into the cylinders, run the engine at 2,000 rpm for a minimum of ten minutes after completing testing to prevent damage to the catalytic converters. Failure to follow this instruction may result in damage to the vehicle.

NOTE:

There is a combustion chamber in the top of each piston. Make sure that the oil is not allowed to run into this chamber.

PINPOINT TEST A : CHECK THE CYLINDER COMPRESSIONS

Only compression testers able to read the higher compression pressures found in diesel engines should be used. Failure to follow this instruction may result in personal injury.

Where possible, compression testing should be carried out on an engine at operating temperature.

TEST CONDITIONS

DETAILS/RESULTS/ACTIONS

A1: CARRY OUT A DRY CYLINDER COMPRESSION TEST

	1 Make sure the parking brake is applied and that the selector lever is in park.
	2 Set the ignition status to OFF .
	3 Remove the starter relay.
	4 Disconnect the starter motor solenoid connector.
	5 Connect a suitable remote starter device to the starter motor solenoid.
	6 Remove the glow plugs.
	7 Install adaptor 303-1131 in place of the glow plug in the first cylinder to be tested.
	8 Connect a suitable compression tester to the adaptor. See warning above.
	9 Using the remote starter device, crank the engine a minimum of five revolutions.
	10 Record the compression figure and the number of revolutions taken to reach it.
	11 Repeat steps 7 - 10 above for the remaining cylinders, cranking the engine for a similar number of revolutions each time.
	12 Compare the compression figures across all the cylinders.

	Are the compression figures within 10% of each other? Yes Unless the compression figures are universally very low (experience will indicate this), check for other causes for the customer complaint. No GO to A2.
--	--

A2: CARRY OUT A WET CYLINDER COMPRESSION TEST

	1 Using a suitable oil can with a flexible spout, introduce a small amount of clean engine oil into the cylinder just before testing, such that the oil is able to run between the piston and the cylinder bore.
	2 Repeat steps 7 - 10 from the test above, introducing oil into each cylinder just before testing.
	3 Compare the compression figures across all the cylinders.
	Is the compression figure higher than the dry test? Yes A higher figure following the introduction of oil may indicate a worn or damaged cylinder bore, piston and/or piston rings. Disassembly would be required to confirm this. No If the compression figure is unaffected by the introduction of oil, but the figure is still less than 90% of the other cylinders, this may indicate a burnt and/or sticking valve, leaking head gasket, etc. Disassembly would be required to confirm this. Clear any DTCs which may have been induced by the test.

NOTE:

Oil consumption will vary, depending on a number of factors. New engines will normally use more oil than 'run-in' engines, although a guideline would be to expect 16,000 Km (10,000 miles) per liter.

PINPOINT TEST B : OIL CONSUMPTION TEST

Oil consumption will vary, depending on a number of factors. New engines will normally use more oil than 'run-in' engines, although a guideline would be to expect 16,000 Km (10,000 miles) per liter.

TEST CONDITIONS

DETAILS/RESULTS/ACTIONS

B1: CHECK FOR EXCESSIVE OIL CONSUMPTION

	1 Start the engine and allow it to run until it reaches normal operating
--	--

	temperature.
	2 Make sure the vehicle is parked on a level surface and set ignition status to OFF .
	3 Allow to settle for at least five minutes.
	4 Check the oil level.
	5 Correct the level, if necessary, and record the reading and mileage in the vehicle history.
	6 Make sure that the owner/driver is aware that a test is being carried out, and that they should not top-up their oil level for the duration of the test, but should check the level every 160-240 Km (100-150 miles).
	7 When the oil level reaches the ADD mark, the customer should bring the vehicle in to be checked.
	8 Top-up the oil to the level at the beginning of the test and record the amount of oil needed to do so, and the mileage covered in the course of the test.
	9 From this, the consumption can be calculated, and a decision made as to whether or not the consumption is considered excessive.
	Is the consumption excessive for the mileage and/or use? Yes Disassembly will be required to check the components indicated in the symptom chart. No No further action is required.

NOTES:

- Check and, if necessary, top-up the engine oil level before beginning this test.
- Where reference is made to 'suitable equipment', this refers to standard workshop equipment. Refer to the operating instructions for your own equipment when performing any tests.

PINPOINT TEST C : CHECK THE ENGINE OIL PRESSURE

Check and, if necessary, top-up the engine oil level before beginning this test.

Where reference is made to 'suitable equipment', this refers to standard workshop equipment. Refer to the operating instructions for your own equipment when performing any tests.

TEST CONDITIONS	DETAILS/RESULTS/ACTIONS
C1: CHECK FOR LOW ENGINE OIL PRESSURE	
	1 Remove the oil pressure sensor.
	2 Connect a suitable oil pressure gauge in place of the oil pressure sensor.
	3 Start the engine and check for leaks at the gauge connection.
	4 Allow the engine to idle and monitor the oil pressure.
	5 Raise the engine speed to 2,500 rpm and monitor the oil pressure.
	Is the oil pressure less than 0.50 bar (7.25 psi) between idle and 2,500 rpm? Yes GO to C2. No GO to C3.
C2: CHECK FOR LOW ENGINE OIL PRESSURE AT ENGINE SPEEDS GREATER THAN 2,500 RPM	
	1 Raise the engine speed to above 2,500 rpm and monitor the oil pressure.
	Is the oil pressure less than 1.0 bar (14.5 psi) at engine speeds greater than 2,500 rpm? Yes Pressure this low may indicate a problem with: oil pump, filtering, clearances within the engine. Check if there are any other indications of engine faults (noise, etc), refer to the symptom chart above. No GO to C3.
C3: CHECK FOR HIGH ENGINE OIL PRESSURE	
	1 Monitor the engine oil pressure at varying engine speeds.
	Does the oil pressure reach 4.0 bar (58 psi)? Yes Pressure this high may indicate a blockage in the lubrication system. If this is not resolved, high oil pressure will lead to engine oil leaks and other failures. No If the oil pressure stays in the band between 0.50 bar (7.25 psi) and 1.0 bar (14.5 psi) this would be considered normal.

DTC INDEX

For a complete list of all Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).